

Release 4 Documentation

Project: Text-to-Network Diagram Translation System

Client: Bosch

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Release Objective

The objective of Release 4 was to finalize the system for presentation and delivery. This release focused on output refinement, scenario-based testing, and diagram generation automation. The goal was to integrate client feedback, validate model behavior, and demonstrate the system's readiness for real-world use cases during the final presentations.

Key Focus Areas

1. **Output Refinement:** Enhanced diagram clarity, added proper annotations, layer separation, and component labeling.
2. **Scenario-Based Testing:** Incorporated realistic use cases, such as multi-region failover and tiered subnet access.
3. **Dataset Mapping:** Finalized the field-level mapping between datasets and diagram components for traceability.
4. **Presentation Preparation:** Prepared comprehensive client and class presentations, walkthroughs, and sample outputs.
5. **Diagram Automation Strategy:** Initiated the design of a pilot automation workflow to generate diagrams from structured input.

Deliverables in Release 4

- Final version of the component-level and relationship-level datasets
- Enhanced AWS diagrams with clear visual structure
- Dataset-to-diagram mapping documentation
- Final client presentation deck
- Class presentation deck and demo materials
- Automation workflow proposal for future deployment

Key Outcomes of Release 4

- Successfully delivered final presentations to client and class.
- Demonstrated refined diagram outputs with improved accuracy and structure.
- Validated hybrid approach combining prompt engineering and fine-tuning.
- Established a foundation for future automation and multi-cloud integration.

Challenges Faced

- Managing detail vs. clarity in diagram outputs.
- Adapting datasets for diverse networking scenarios.
- Balancing final presentation preparation with ongoing output validation.

Next Steps

- Explore integration of diagram generation in CI/CD pipelines.
- Expand dataset to support Azure and GCP.
- Build a visual prompt builder interface.
- Publish project as a reusable AI-based infrastructure design assistant.